CS 6515: Graduate Algorithms

Lecture 11: Why Strong Duality Holds

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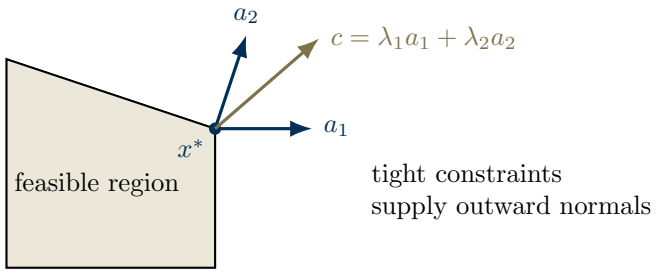
Today's question. Weak duality is an easy calculation. Why must an exact upper-bound certificate exist whenever the primal has a finite optimum? Our proof combines a geometric certificate of infeasibility (Farkas' lemma) with the active-constraint viewpoint.

1 The proof plan

Recall

$$\begin{array}{lllll} (P) & \max c^\top x & Ax \le b, & x \ge 0, \\ (D) & \min b^\top y & A^\top y \ge c, & y \ge 0. \end{array}$$

Let x^* be an optimal primal solution. A constraint $a^\top x \le d$ that is tight at x^* forbids moving in a direction v with $a^\top v > 0$. An inactive constraint has positive slack, so it does not prevent a sufficiently small move.



At an optimum, the objective direction is a nonnegative combination of the active outward normals.

We will justify the picture in three steps:

1. Either c is a nonnegative combination of the active normals, or a separating direction points toward a better feasible solution.
2. A better feasible solution contradicts optimality.
3. The nonnegative coefficients form a dual solution with the same objective value as x^*.

Important bookkeeping. The active normals include those of $-x_i \le 0$. For instance, $\max(-x)$ subject to $0 \le x \le 1$ is optimized at $x = 0$; the relevant tight normal is -1, not the inactive upper bound's normal $+1$.

2 Farkas' lemma: certificates of infeasibility

Given columns $a_1, \ldots, a_k \in \mathbb{R}^d$, their nonnegative combinations form a cone

$$C = \left\{ \sum_{j=1}^{k} \lambda_j a_j : \lambda_j \geq 0 \right\}.$$

The question $M\lambda = p,\ \lambda \geq 0$, with $M = [a_1\ \cdots\ a_k]$, asks whether p belongs to this cone.

Theorem (Farkas' lemma). For any real matrix M and vector p, exactly one of the following systems has a solution:

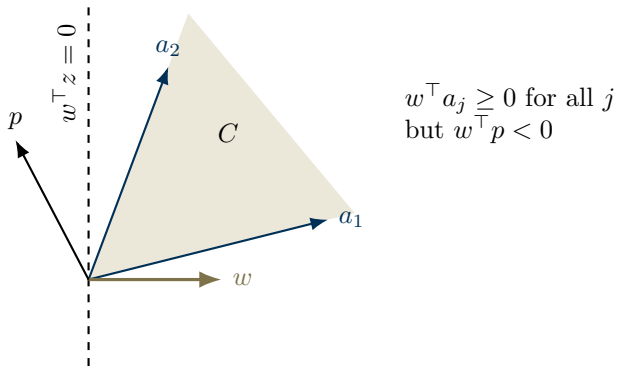
$$\begin{aligned} &\text{(I)} \quad M\lambda = p, \quad \lambda \geq 0; \\ &\text{(II)} \quad M^\top w \geq 0, \quad p^\top w < 0. \end{aligned}$$

The vector w in (II) is unrestricted in sign.

Why they cannot both hold. If they did, then

$$p^\top w = (M\lambda)^\top w = \lambda^\top M^\top w \geq 0,$$

contradicting $p^\top w < 0$.



Example. Take $a_1 = (1, 0)$, $a_2 = (1, 1)$, and $p = (0, 1)$. No nonnegative combination equals p. The certificate $w = (1, -1)$ satisfies $a_1^\top w = 1$, $a_2^\top w = 0$, and $p^\top w = -1$.

The only nontrivial part of Farkas' lemma is that a separating certificate *always* exists when $p \notin C$.

3 The geometric reason: separation

Theorem (separating a point from a closed convex set). Let $C \subseteq \mathbb{R}^d$ be nonempty, closed, and convex, and let $p \notin C$. There are a vector v and a point $q \in C$ such that

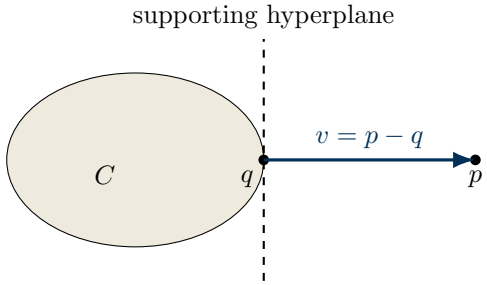
$$v^\top p > v^\top q \geq v^\top z \qquad \text{for every } z \in C.$$

Proof. Choose a closest point $q \in C$ to p, and set $v = p - q$. A closest point exists: after fixing any candidate in C, we can restrict to its closed distance ball around p, which is compact.

For every $z \in C$, convexity keeps $q + \theta(z - q)$ feasible for $0 \leq \theta \leq 1$. If $v^\top(z - q) > 0$, then for sufficiently small $\theta > 0$,

$$\begin{aligned} \|p - (q + \theta(z - q))\|_2^2 &= \|p - q\|_2^2 - 2\theta v^\top(z - q) + \theta^2\|z - q\|_2^2 \\ &< \|p - q\|_2^2, \end{aligned}$$

a contradiction. Thus $v^\top z \leq v^\top q$. Finally $v^\top p - v^\top q = \|p - q\|_2^2 > 0$.



3.1 Finish the proof of Farkas' lemma

We use one standard geometric fact: a cone generated by finitely many vectors is closed (as well as convex). If $p \notin C$, apply separation. Since $0 \in C$, the separating upper bound is nonnegative. Since C contains every nonnegative multiple of each a_j, a finite upper bound requires $v^\top a_j \leq 0$ for every j. Thus the largest value of $v^\top z$ on C is 0, and $v^\top p > 0$. Set $w = -v$: then $M^\top w \geq 0$ and $p^\top w < 0$, as required.

What to remember. Infeasibility of a nonnegative linear combination has a short *linear* witness. The geometric proof explains why the witness exists.

4 Strong duality: the active-constraint argument

Write *all* primal inequalities as

$$a_r^\top x \le d_r \qquad (r = 1, \dots, m+n),$$

where the last n rows are $-x_i \le 0$. Let S be the rows tight at an optimal primal solution x^*. We claim

$$c = \sum_{r \in S} \lambda_r a_r \qquad \text{for some } \lambda_r \ge 0. \tag{1}$$

4.1 If the claim fails, Farkas gives an improving direction

Apply Farkas' lemma to the matrix whose columns are a_r, $r \in S$, with target c. If (1) is impossible, there is w with $a_r^\top w \ge 0$ for all $r \in S$ and $c^\top w < 0$. Let $v = -w$. Then

$$a_r^\top v \le 0 \quad (r \in S), \qquad c^\top v > 0.$$

Consider $x^* + \varepsilon v$:

- Tight constraints stay feasible because $a_r^\top (x^* + \varepsilon v) \le d_r$.
- An inactive constraint has slack $\delta_r = d_r - a_r^\top x^* > 0$. If $a_r^\top v > 0$, choose $\varepsilon \le \delta_r / (a_r^\top v)$; otherwise it imposes no restriction.

There are finitely many rows, so a sufficiently small $\varepsilon > 0$ satisfies all of them, including the nonnegativity rows. But its objective increases by $\varepsilon c^\top v > 0$, a contradiction. This proves (1).

4.2 Turn the coefficients into a dual solution

Give inactive rows coefficient zero. Let $y_j \ge 0$ be the coefficients of the original m rows, and $z_i \ge 0$ the coefficients of $-x_i \le 0$. Then

$$c = A^\top y - z \quad \Longrightarrow \quad A^\top y \ge c.$$

So y is dual-feasible. Every row with positive coefficient is tight; the bound rows have right-hand side 0. Therefore

$$c^\top x^* = \sum_r \lambda_r a_r^\top x^* = \sum_r \lambda_r d_r = b^\top y.$$

Weak duality now proves that y is dual-optimal and the two optimal values coincide. Notice that x^* did not need to be a vertex.

5 What these three LP lectures give us

1. **Modeling.** Real variables and linear constraints express resource allocation and graph problems.
2. **Algorithms.** Polynomial-size rational LPs can be solved in polynomial time in their encoding length.
3. **Certificates.** A feasible solution and a matching-value dual solution prove optimality.
4. **Geometry.** At an optimum, tight constraints block every improving direction. Farkas' lemma converts this fact into multipliers.
5. **Rounding.** A fractional LP optimum can guide the search for an integral solution; max-flow/min-cut is an exact example.

5.1 Bridge to convex optimization

For an LP, the objective has a fixed direction c. For a differentiable nonlinear objective f, the gradient $\nabla f(x)$ gives a local linear approximation:

$$f(x+h) \approx f(x) + \nabla f(x)^\top h.$$

Convexity will make that local information globally useful.

We will next *minimize* a convex function over a convex set. A maximization LP fits this viewpoint by minimizing the convex linear function $-c^\top x$. We will then move beyond linear objectives to examples such as least squares.

5.2 Checks for understanding

Exercise. For $\max(-x)$ subject to $x \le 1,\ x \ge 0$, write its dual and find an optimal primal-dual pair. Which nonnegativity normal supplies the objective direction in the strong-duality proof?

Exercise. Find a feasible LP with a unique optimum for which a tight constraint has zero dual multiplier. Why does this not contradict complementary slackness?

Exercise. Using only weak duality, show that a feasible pair with $b^\top y - c^\top x \le \varepsilon$ certifies $\mathrm{OPT} - c^\top x \le \varepsilon$.